

Stockout-Aware Demand Forecasting and Newsvendor Ordering on FreshRetailNet-50K

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Abstract—Recorded sales understate demand whenever a product sells out, so forecasts and orders fitted to sales inherit the stocking policy. We study demand forecasting and newsvendor ordering on FreshRetailNet-50K, a fresh-retail panel with hourly stockout flags. Days without a flagged stockout give an exact, though selected, proxy for demand. We censor these days with three simulated stockout mechanisms and score every forecast and order against the same uncensored outcomes, so models trained on differently censored histories are compared on identical rows. Earlier evaluations on this dataset score demand recovery and forecast accuracy, or profit measured on censored sales; we score ordering cost against an uncensored proxy and measure transfer between censoring mechanisms. In a 5,000-series run locked before the final week was scored, we compare nine ordering policies under three cost ratios; none wins under every ratio. When a shortage costs three times as much as a leftover unit, LightGBM demand quantiles with pooled conformal calibration are the cheapest full-sample policy in all nine train/test cells; when leftovers cost more, a raw-sales LightGBM with residual quantiles is cheapest in all nine. Training on a different mechanism raises the calibrated policy’s mean cost by at most 0.0472 normalized units (95% cluster-bootstrap interval 0.0391–0.0554), the smallest worst case among the policies that read the evaluation history. For real censored days, where demand is unobserved, we report only a lower bound on misses of the 90% quantile forecast.

Index Terms—censored demand, stockouts, demand forecasting, quantile regression, conformal calibration, newsvendor, fresh retail

I. INTRODUCTION

Retailers record sales, not demand. When a product sells out, the record stops at the stock that was available and the unmet part of demand leaves no trace [1]. A model fitted to such records learns the stocking policy along with customer behavior, and its errors carry into the next order. Fresh retail makes the cost immediate: tomorrow’s order trades lost sales against perishable surplus, and a forecast that tracks recorded sales closely can still lead to a poor order if its history was censored [2], [3]. A lower forecast error therefore does not by itself show that one ordering policy is better than another. The comparison has to use the same outcome, the same cost ratio, and the same scored rows.

FreshRetailNet-50K [4], [5] makes this problem measurable because it records an hourly stock-status flag next to hourly

sales. A series-day with no flagged stockout between 06:00 and 21:59 is a *donor day*, and its recorded sales over that window are an exact proxy for demand, provided the flags are correct. Donor days are not a random sample: well-stocked and slow-moving items qualify more often, and the proxy says nothing about demand on days that were actually censored. Within that limit, donor days support a controlled experiment. The dataset report uses non-stockout periods in this way to score demand recovery [4]; we carry the idea from hourly recovery to next-day orders. Three simulated stockout mechanisms censor the histories of eligible donor days in different ways, features and forecasts are rebuilt from each censored history, and every forecast and order is scored against the same uncensored proxy. A model trained under one mechanism can thus be evaluated under another on identical rows.

The evidence comes from three saved executions of one notebook pipeline (Fig. 1). Two development runs prepare the data, simulate censoring, and tune the models: run S on a stratified sample of 5,000 series and run P on all 50,000 series. Run E, the end-to-end run, repeats these steps on its own 5,000-series sample and continues through neural and censored-inventory baselines, conformal calibration, ordering, a protocol lock, and a single evaluation of the final week. Only run E feeds the final evaluation.

Our contribution is an evaluation design and the evidence it produces, not a new forecasting architecture. Earlier work on this dataset scores demand recovery and forecast accuracy [4] or measures profit on censored sales [6]. We add the following:

- we score ordering cost, not only forecast error, against an uncensored proxy, so that nine policies are compared on identical rows with an identified outcome;
- we vary how the training history was censored and measure the cost of transfer between mechanisms, with paired cluster-bootstrap intervals;
- we show that the ranking of calibrated demand quantiles and a raw-sales model reverses with the cost ratio in every train/test cell, and we bound the miss rate of the 90% quantile forecast (q90) on real censored days, where no demand label exists.

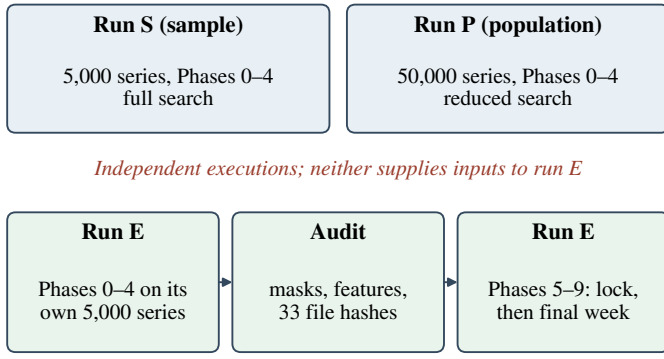


Fig. 1. The three saved executions. Only run E’s own Phase 0–4 stage feeds its prerequisite audit and the locked final evaluation; runs S and P are independent development evidence.

The code, saved run outputs, and paper sources are public.¹

A. Related work

Censored demand and the newsvendor. Parametric methods fit a demand distribution to censored observations [2], and nonparametric policies learn order levels from censored sales with the Kaplan–Meier product-limit estimator [3], [7]; our Kaplan–Meier baseline follows the second idea with pooled support rules. Theory characterizes how censoring changes what can be learned about the optimal order [8], data-driven newsvendor methods have been adapted to censored samples [9], and treating sales as demand can degrade badly as censored data accumulate [10]; our raw-sales baselines test that heuristic empirically. Feature-based and neural newsvendor methods learn orders directly from covariates [11]–[13]; under linear costs the optimal order is a quantile, which ties ordering to quantile regression [14].

Forecasters and calibration. Our forecasters are gradient-boosted trees [15], TimesNet [16] for recovering masked hours, and the Temporal Fusion Transformer (TFT) [17] for next-day forecasts. Tree ensembles are a strong default for retail sales: LightGBM was used by practically all of the top 50 entries in the M5 accuracy competition [18], and simple models often match transformer forecasters [19]. Calibration applies pooled split-conformal corrections in the spirit of conformalized quantile regression [20], a sequential variant as in adaptive conformal inference [21], and monotone rearrangement to remove quantile crossing [22]; conformal calibration of the critical quantile has also been proposed for the feature-based newsvendor [23]. Coverage guarantees rest on exchangeability or on explicit bounds for departures from it [24]. Real stockout-censored days meet neither condition, so we measure coverage only where the donor proxy exists.

Evaluation on stockout-annotated data. The dataset report [4] introduces the stockout annotations, a two-stage recovery-then-forecast baseline, and a controlled missing-not-at-random simulation that masks synthetic regions of non-stockout periods to score recovery; we reproduced its official code [25] only in

part (Section IV-A). Azar [6] builds a leakage-free LightGBM pipeline on all 50,000 series and selects among newsvendor policies on a validation fold, but scores profit on censored sales, whose bias relative to demand is not identified. Elsewhere, stockout-aware evaluation excludes out-of-stock days from training losses and accuracy metrics [26], and a global gradient-boosted model with stockout features feeds a cost-aware ordering rule [27]. None of these scores orders against an uncensored outcome while varying the censoring mechanism, which is the gap our design addresses.

II. DATA AND EXPERIMENTAL PROTOCOL

A. Data and outcome

The local training file has 4,500,000 series-day rows and the evaluation file 350,000. A series is a (city, store, product) triple, and the training panel contains 50,000 series. The local training file holds 865 distinct product identifiers, the number given on the current dataset card [5]; the pinned dataset report states 863 stock-keeping units (SKUs) [4]. We use release 1.0 (CC BY 4.0); its upstream revision and our download date were not recorded, so the SHA-256 hashes in the availability statement identify the files.

The outcome is the sum of recorded hourly sales from 06:00 to 21:59, the window a next-day order has to cover, rather than the full-day `sale_amount`. A series-day is a donor if its stock-status flags $s_{j,t,h}$ sum to zero over that window; flags outside the window do not matter:

$$I_{j,t}^{\text{donor}} = \mathbf{1} \left\{ \sum_{h=6}^{21} s_{j,t,h} = 0 \right\}, \quad (1)$$

$$D_{j,t}^{\text{proxy}} = \sum_{h=6}^{21} y_{j,t,h}^{\text{sale}} \quad \text{if } I_{j,t}^{\text{donor}} = 1.$$

The selection of donor days is measurable: in run S, fit-period donor days had a pre-day two-week mean of recorded sales 0.91 times that of the other days and fell less often on promotion days (0.35 against 0.39). Real final-week rows with a flagged stockout have no demand label and are never scored for error or cost (see Section V-E).

B. Pipeline and chronological protocol

The pipeline has ten phases: the data contract and sampling (Phase 0), audit and past-only features (1), simulated censoring (2), conventional baselines (3), LightGBM demand quantiles (4), TimesNet and TFT (5), Kaplan–Meier (6), conformal calibration (7), ordering policies and the protocol lock (8), and final evaluation (9). Table I gives the fixed chronological split. Sampling tiers and the initial simulator rules use fit-period data only, hyperparameters are chosen on the tuning window, and the calibration window is used for calibration alone. The 5,000-series samples are stratified by first-level category, sales tier, and stockout-rate tier.

Features for a target date use only earlier information: target-day stockout flags, sales, weather, and promotion values are excluded, and lag and rolling features are shifted by at least one day; randomizing every target-day field leaves the features

¹<https://github.com/Research-maniacs/stockout-aware-demand-forecasting>

TABLE I
CHRONOLOGICAL SPLIT SHARED BY ALL RUNS

Role	Days	Dates	Use
Model fit	1–60	2024-03-28 to 2024-05-26	model fitting
Tuning	61–75	2024-05-27 to 2024-06-10	model selection
Calibration	76–90	2024-06-11 to 2024-06-25	calibration only
Final	91–97	2024-06-26 to 2024-07-02	locked evaluation

unchanged. Until Phase 4 the evaluation file was read only for its schema, identifiers, and dates. Run E wrote its protocol lock in Phase 8, before any final-week outcome was scored, and the final week has been used once, for the evaluation reported here.

C. Simulated censoring mechanisms

Each mechanism replaces the visible history of eligible donor days, those with enough past data, by a simulated censored history; real censored days keep their recorded values. The opening stock of series j on day t under mechanism m is

$$Q_{j,t}^{(m)} = \max\left\{\mu_{j,t}^{\text{past}} k_{j,t}^{(m)} \exp(\sigma_m z_{j,t}), q_{\text{floor}}\right\}, \quad (2)$$

where $\mu_{j,t}^{\text{past}}$ is the mean recorded window sales over the preceding two weeks, $k_{j,t}^{(m)}$ a stock multiplier fitted on fit-period stockout patterns, $z_{j,t}$ a seeded standard normal draw that all mechanisms share, and q_{floor} a small floor. Visible hourly sales are the lesser of proxy sales and remaining stock. Mechanism A (ordinary depletion) may add one refill of half the opening stock one to three hours after the first simulated stockout, with a probability estimated from real within-day recoveries. Mechanism B (promotion-sensitive) follows A but multiplies the stock factor by 0.6 on days that were promotion days in the record, so those days deplete earlier; the promotion flag drives the simulator and is not a forecasting feature. Mechanism C (sustained scarcity) halves A's multiplier, uses a fixed noise scale, and never refills (Fig. 2).

Within a transfer cell, defined by a training mechanism and an evaluation history, every forecast is scored against the same proxy outcome, so cost differences are paired. Across mechanisms, the masking rate and the distribution of reconstructed features change by design: B probes promotion-linked depletion and C persistent shortage.

D. Models, calibration, and ordering

The conventional baselines are a seasonal-naive rule and LightGBM [15] point and quantile models trained on visible sales. The demand-quantile family trains LightGBM quantile models on donor-proxy targets at $\tau \in \{0.25, 0.50, 0.75, 0.90\}$, with one set per mechanism. On a declared deep subset of series, TimesNet recovers artificially hidden hours and TFT forecasts the next day either from the visible history (direct) or from the TimesNet-recovered history. The Kaplan–Meier policy estimates an order percentile from censored sales within pools that fall back from category by sales tier to category, management group, and a global pool when support is too small.

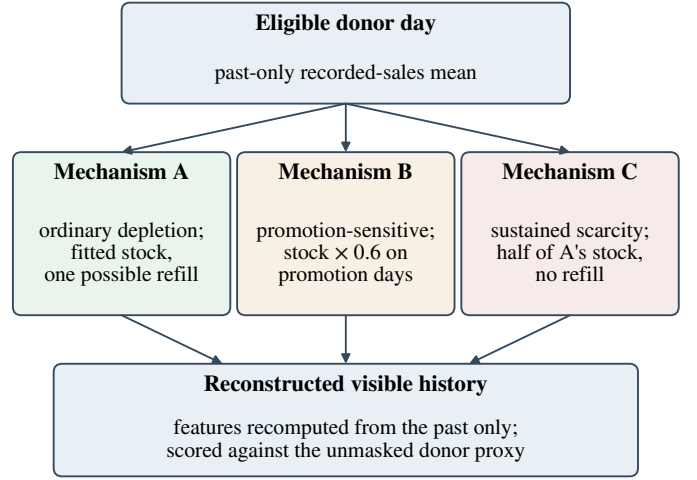


Fig. 2. The three censoring mechanisms act on eligible donor days. Each produces its own visible history, from which past-only features and forecasts are rebuilt; real censored days keep their recorded values.

Pooled conformal calibration shifts each quantile by a residual quantile computed on calibration-window donors in the same pool hierarchy, with a minimum pool size of 200, and then sorts the corrected quantiles of each row. The adaptive variant updates its target miss rate after each donor day, as in adaptive conformal inference [21], and serves as a sensitivity analysis. Point forecasters (seasonal naive, raw-sales LightGBM, and TFT) become quantile forecasts by adding the pooled quantile of their calibration residuals.

With a cost c_u per unit short and c_o per unit left over, the newsvendor order is the $\beta = c_u/(c_u + c_o)$ quantile of the forecast distribution. The cost of a row is

$$C(Q, D^{\text{proxy}}) = c_o \max(Q - D^{\text{proxy}}, 0) + c_u \max(D^{\text{proxy}} - Q, 0), \quad (3)$$

evaluated against the unmasked donor proxy under a one-day shelf life. The locked settings are $(c_u, c_o, \beta) = (1, 3, 0.25)$, $(1, 1, 0.50)$, and $(3, 1, 0.75)$, which we call excess stock more costly, equal costs, and shortage more costly. Costs are in normalized sales units.

E. Metrics

For a scoring population $\mathcal{I}$ of donor rows, we report the weighted absolute percentage error of the q50 forecast, the empirical coverage of q_τ , and the pinball loss [14]:

$$\text{WAPE} = \frac{\sum_{i \in \mathcal{I}} |D_i^{\text{proxy}} - \hat{D}_i|}{\sum_{i \in \mathcal{I}} D_i^{\text{proxy}}}, \quad (4)$$

$$\hat{C}_\tau = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \mathbf{1}\{D_i^{\text{proxy}} \leq q_{\tau,i}\}, \quad (5)$$

$$\hat{P}_\tau = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} (D_i^{\text{proxy}} - q_{\tau,i})(\tau - \mathbf{1}\{D_i^{\text{proxy}} < q_{\tau,i}\}). \quad (6)$$

The development files store WAPE as a ratio; the final forecast file stores it as a percentage. Coverage on calibration rows is only a diagnostic, because the same rows set the correction.

TABLE II
DEVELOPMENT RUNS, PHASES 0–4 (TUNING WINDOW; WAPE AS A RATIO)

Measure	Run S	Run P
Selected series	5,000	50,000
Selected series-days	450,000	4,500,000
Hyperparameter search	full	reduced
Fit donor days	163,250	1,630,675
Tuning donor days	45,097	450,872
Calibration donor days	42,773	426,447
Overall donor share	0.5580	0.5573
Training rows per mechanism	144,078	1,440,065
q50 tuning WAPE, A	0.3334	0.3124
q50 tuning WAPE, B	0.3361	0.3144
q50 tuning WAPE, C	0.3358	0.3161
Runtime (s)	1,194.9	4,210.4
Peak memory (GB)	1.292	6.110

On donor rows, forecast error, coverage, pinball loss, and order cost answer different questions, and we do not rank methods across them.

For a real final-week row, let S_{rec} be the recorded window sales and U_{90} the q90 forecast. Demand on a censored row is at least S_{rec} , so every row with $S_{\text{rec}} > U_{90}$ is a certain miss, whereas a row with $S_{\text{rec}} \leq U_{90}$ may still hide one. The definite-miss rate

$$L_{\text{miss}} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{S_{\text{rec},i} > U_{90,i}\} \quad (7)$$

is therefore a lower bound on the real q90 miss rate. It is neither a coverage estimate nor an estimate of censored demand.

III. DEVELOPMENT RUNS AND REPRODUCIBILITY

Table II compares the two development runs. Run S searched hyperparameters fully on 5,000 series; run P used a reduced search on all 50,000 series. The overall donor share is almost the same (0.5580 and 0.5573), so the stratified sample preserves donor availability. Run P has a lower q50 tuning WAPE for all three mechanisms and a longer runtime, but the runs differ in search mode as well as size, so neither difference can be attributed to scale and the runtimes are not a scaling benchmark. Fit-period donor days outnumber training rows because the first week of history only feeds lag features.

The simulator summaries agree between the two runs to within 0.001. The masked share, the fraction of eligible donor days with at least one simulated stockout hour in the window, is 0.460, 0.566, and 0.832 for A, B, and C in run S (0.459, 0.567, and 0.832 in run P), and the share of proxy sales left visible after simulation is 0.803, 0.718, and 0.474 (0.804, 0.717, and 0.474). The sample is therefore representative for these summaries; this does not show that the joint feature distribution, or any downstream metric, is the same at both scales.

Run E did not reuse run S. It executed its own Phase 0–4 stage, with the full hyperparameter search, in the same kernel as its later phases. Its fingerprints of the selected series, the chronological split, and the three simulator masks equal those of run S, and so do its donor counts and its q50 tuning WAPEs at full stored precision (0.333440, 0.336140, and 0.335759

for A, B, and C). This is consistent with a deterministic pipeline applied to the same data, seed, and sampling rule. The two runs still differ in timestamps, pandas version (3.0.2 in run S, 2.3.3 in run E), and file hashes, so we treat them as separate evidence. Within run E, one Phase 0 check stayed open because it compares fingerprints with an earlier execution in the same output folder, and the first execution has none; the comparison with run S above is the one it would have made. Before continuing, a prerequisite audit re-derived the simulator masks and feature matrices from the raw panel, required exact agreement, and verified the hashes of 33 predecessor files. Run E’s later phases ran in the reduced mode, which uses 200 bootstrap replicates. No run hashed the notebook source it executed, so the current notebook files cannot prove which code produced a given result.

IV. MODELS, CALIBRATION, AND THE PROTOCOL LOCK

All results from here on come from run E and its 5,000 series. Its prerequisite audit passed 10 checks, which also confirmed that proxy labels exist only on donor days and match a recomputation and that the calibration and final-week days were still unused.

A. Neural models as budget-limited references

The neural models ran on CPU, with two logical CPUs, 8.42 GB of RAM, and four training epochs per model, on a declared subset of 600 series. This budget makes them reference points rather than tuned competitors, so we compare them only with each other and within the subset, and we draw no conclusion about neural versus tree-based forecasters; tree ensembles are already a strong default for retail sales [18]. TimesNet recovers artificially hidden donor hours; across the three mechanisms it scored 290,397 hidden hours with WAPE 0.917 to 0.934. TFT forecasts the next day’s window total on 5,407 matched tuning donor days, and its WAPE falls from 0.451–0.453 with the visible history to 0.413–0.415 with the TimesNet-recovered history. The two tasks differ in target, unit, and denominator, and neither WAPE is comparable with the Phase 4 q50 WAPE on the full sample. The final TFT routes reuse 6 trained models; the 24 frozen prediction pipelines combine the direct and recovered variants with the three training mechanisms and four evaluation histories (A, B, C, and the real history).

Our reproduction of the official baseline [25] is partial. The as-documented TimesNet run failed on a blocked network download and completed only after a recorded patch that reads the verified local Parquet files; the upstream TFT training finished, but its prediction step was not run. The task-aligned TimesNet and TFT implementations used in this paper passed their own stage checks.

B. Kaplan–Meier baseline and calibration

The Kaplan–Meier policy passed all 8 of its unit tests and built 207 pools, with support thresholds chosen on tuning donors; when a pool cannot support the requested percentile,

the order is left unestimated rather than borrowed from another policy.

Calibration used 513,276 calibration-window residuals, pooled over mechanisms and quantile levels; 0.061 of the calibration rows were corrected from a pool coarser than category by sales tier. Sorting each row's corrected quantiles removed every remaining crossing, which guarantees ordered quantiles, not calibrated ones.

C. Ordering policies and the protocol lock

Nine policies turn forecasts into orders (Table III). All 10 hand-computed checks of the order and cost arithmetic passed. The protocol lock written in Phase 8 then fixed the policy names, cost settings, model and calibration files, planned comparisons, and bootstrap settings before Phase 9 read any final-week outcome. The final week has been scored once under this lock, and any revised protocol applied to the same week would be exploratory.

V. LOCKED FINAL-WEEK EVALUATION

The final evaluation scores donor rows only. Each of the nine transfer cells, one per pair of training mechanism and evaluation history, holds the same 20,641 final donor series-dates, and the six full-sample policies other than Kaplan–Meier are scored on all of them. Kaplan–Meier scores only rows with an estimable order, between 6,012 and 20,641 per cell, with the fewest in the C-trained cells when shortage is more costly. The two neural policies score the 2,510 final donor rows of the deep subset, and no predictions are imputed for the remaining 4,400 series. Tables therefore state each policy's population and row count.

A. Policy comparison

Table III gives the mean costs of the A-trained models on the A history under the three cost settings. The best full-sample policy depends on the cost ratio. On the A history, the raw-sales LightGBM with residual quantiles has the lowest cost when excess stock is more costly and, narrowly, at equal costs (0.3868 against 0.3891 for pooled conformal); the pooled-conformal policy has the lowest cost when shortage is more costly. The B history gives the same ranking. On the C history the pooled-conformal policy is already the cheaper of the two at equal costs (0.4094 against 0.4411), so averaged over the three matched histories it is cheaper at equal costs (0.3984 against 0.4075), as in Tables IV and VIII; the paired interval of that average difference contains zero. The Kaplan–Meier percentile is the most expensive full-sample policy in each setting of the table. Within the deep subset, TFT on the TimesNet-recovered history costs less than direct TFT in every matched history and cost setting, a comparison internal to that subset.

Table IV reports the paired contrasts that the lock pre-specified for the pooled-conformal policy, pooled over the three matched histories. Calibration lowers cost relative to the uncalibrated demand quantiles in all three settings, and the pooled-conformal policy is cheaper than the seasonal-naive and Kaplan–Meier policies throughout. Against the raw-sales

LightGBM with residual quantiles the sign depends on the cost ratio: the pooled-conformal policy costs more when excess stock is more costly, the equal-cost interval contains zero, and it costs less when shortage is more costly.

B. Service, surplus, and coverage

Table V shows the pooled-conformal policy in each matched history. Moving from $\beta = 0.25$ to $\beta = 0.75$ raises the fill rate on the A history from 0.614 to 0.891 and the leftover share from 0.056 to 0.226. The critical fractile moves the order along the trade-off between service and surplus without removing it. Mean costs are not comparable across settings because the cost weights differ. The C history is the most expensive in every setting.

Pooled calibration also moves the empirical q90 coverage of the 20,641 matched final donor rows toward the nominal 0.90 in every history, from 0.798–0.828 to 0.879–0.887, and lowers the q90 pinball loss from 0.119–0.131 to 0.109–0.114. Coverage stays below nominal. It is measured on selected complete days after the calibration window, so it does not show that exchangeability holds for censored days or would survive a change of stocking policy.

C. Mechanism transfer

Table VI fixes the pooled-conformal policy at $\beta = 0.75$ and varies the history used for training and calibration. The transfer gap

$$\Delta_{m \rightarrow h} = \bar{C}_{m \rightarrow h}(I) - \bar{C}_{m \rightarrow m}(I) \quad (8)$$

compares the mean cost on an unseen evaluation history h with the matched cost on the same rows I . The largest gap arises for models trained on A and evaluated on C: 0.7303 against 0.6831, a gap of +0.0472. Training on B and evaluating on C also raises the cost. The C-trained models, in contrast, cost less on the A and B histories than on their own, consistent with the C history being the hardest to forecast from (Table V). A negative gap therefore signals an easier evaluation history rather than better transfer, and gaps combine a change of model with a change of difficulty.

All intervals are 95% percentile intervals from 200 cluster-bootstrap replicates [28]. Each replicate resamples the 866 city–store clusters with replacement and keeps every row of a drawn cluster, and both sides of a contrast use the same draws. Dates are not resampled separately and no multiplicity adjustment is applied. With 200 replicates the tail percentiles are approximate, so we read the intervals as descriptive and make no claims of statistical significance.

D. Robustness across cells, dates, and series

The reversal with the cost ratio is not specific to the matched histories. Over all nine train/test cells, the raw-sales LightGBM with residual quantiles is the cheapest full-sample policy in every cell when excess stock is more costly, with the pooled-conformal policy 0.0007 to 0.0323 more expensive; when shortage is more costly, the pooled-conformal policy is the cheapest in every cell, 0.0441 to 0.1298 below the raw-sales policy. At equal costs the cells split: four go to each of the

TABLE III
MEAN NEWSVENDOR COST OF EVERY LOCKED POLICY FOR A-TRAINED MODELS ON THE A HISTORY (LOWEST COST PER PANEL AND SETTING IN BOLD)

Policy	Population	n	Mean cost		
			$\beta = 0.25$	$\beta = 0.50$	$\beta = 0.75$
<i>(a) Full-sample exact-donor candidates</i>					
Seasonal naive + residual quantile	Full sample	20,641	0.6759	0.4430	0.7853
Raw-sales LightGBM + residual quantile	Full sample	20,641	0.5448	0.3868	0.7366
Raw-sales quantile LightGBM	Full sample	20,641	0.5749	0.3999	0.7503
Demand-quantile LightGBM (uncalibrated)	Full sample	20,641	0.5816	0.3946	0.7152
Demand-quantile LightGBM + pooled conformal	Full sample	20,641	0.5766	0.3891	0.6831
Kaplan–Meier order percentile	Full sample	20,641	0.8404	0.6520	1.3878
Demand-quantile + adaptive calibration (sensitivity)	Full sample	20,641	0.5815	0.3932	0.6963
<i>(b) Declared deep subset of 600 series (separate population)</i>					
TFT (direct) + residual quantile	Deep subset	2,510	0.6892	0.4949	0.9483
TimesNet→TFT + residual quantile	Deep subset	2,510	0.6586	0.4687	0.8759

TABLE IV
PRE-SPECIFIED PAIRED CONTRASTS FOR THE POOLED-CONFORMAL POLICY OVER THE THREE MATCHED HISTORIES (NEGATIVE VALUES FAVOR POOLED CONFORMAL)

Pooled conformal minus	Cost setting	Population	n	Difference	95% percentile interval
Uncalibrated demand quantile	Excess stock more costly	Matched full sample	61,923	−0.0052	[−0.0068, −0.0034]
Uncalibrated demand quantile	Equal costs	Matched full sample	61,923	−0.0077	[−0.0096, −0.0062]
Uncalibrated demand quantile	Shortage more costly	Matched full sample	61,923	−0.0420	[−0.0483, −0.0365]
Raw-sales LightGBM + residual	Excess stock more costly	Matched full sample	61,923	+0.0208	[0.0016, 0.0432]
Raw-sales LightGBM + residual	Equal costs	Matched full sample	61,923	−0.0091	[−0.0204, 0.0052]
Raw-sales LightGBM + residual	Shortage more costly	Matched full sample	61,923	−0.0790	[−0.0946, −0.0615]
Seasonal naive + residual	Excess stock more costly	Matched full sample	61,923	−0.1213	[−0.1465, −0.0876]
Seasonal naive + residual	Equal costs	Matched full sample	61,923	−0.0694	[−0.0872, −0.0439]
Seasonal naive + residual	Shortage more costly	Matched full sample	61,923	−0.1376	[−0.1723, −0.0902]
Kaplan–Meier percentile	Excess stock more costly	KM-supported rows	61,523	−0.4043	[−0.4276, −0.3824]
Kaplan–Meier percentile	Equal costs	KM-supported rows	58,183	−0.3223	[−0.3429, −0.3046]
Kaplan–Meier percentile	Shortage more costly	KM-supported rows	47,294	−0.7475	[−0.8175, −0.6812]

TABLE V
POOLED-CONFORMAL POLICY IN THE MATCHED HISTORIES

History	Population	n	Mean cost	Fill rate	Leftover
<i>Excess stock more costly ($\beta = 0.25$)</i>					
A	Full sample	20,641	0.5766	0.614	0.056
B	Full sample	20,641	0.5825	0.608	0.056
C	Full sample	20,641	0.5978	0.589	0.055
<i>Equal costs ($\beta = 0.50$)</i>					
A	Full sample	20,641	0.3891	0.775	0.124
B	Full sample	20,641	0.3967	0.769	0.125
C	Full sample	20,641	0.4094	0.753	0.122
<i>Shortage more costly ($\beta = 0.75$)</i>					
A	Full sample	20,641	0.6831	0.891	0.226
B	Full sample	20,641	0.6940	0.888	0.228
C	Full sample	20,641	0.7248	0.878	0.227

two policies and one to the uncalibrated demand quantiles. Over the three cost settings, the largest transfer gap of the pooled-conformal policy is +0.0472, the smallest worst case among the six policies whose forecasts read the evaluation history; the raw-sales quantile LightGBM reaches +0.2258. Kaplan–Meier orders do not read that history, so its gaps are zero by construction. Over the seven final dates, with 2,507 to 3,385 donor rows each, the C history is the most expensive

on every date in all three cost settings, and the daily q90 coverage of the pooled-conformal models ranges from 0.839 to 0.907. Under-coverage concentrates in series with at most ten complete days in the fit period (q90 coverage 0.752 to 0.769 over 532 rows) and on promotion days (0.854 to 0.865). These breakdowns are descriptive outputs of the same locked run, not pre-specified comparisons, and carry no intervals.

E. Real recorded histories

The q90 forecasts of the three pooled-conformal models were applied to the same 35,000 real final-week rows, 105,000 forecast rows in all, including rows with flagged stockouts. Table VII gives the definite-miss lower bound of (7). A smaller bound can come from a higher forecast, so it does not by itself indicate better calibration. Across strata, the bound is highest for rows with 1–4 recorded stockout hours and lowest for 9–16 hours. Heavy censoring weakens the test: when a product sells out early, recorded sales can fall far below demand and seldom exceed the forecast even when demand does. Only the complete-day stratum has an exact proxy.

VI. DISCUSSION AND LIMITATIONS

The reversal with the cost ratio follows the direction of the censoring bias. Sales understate demand, which costs little when the order targets a low quantile and much when it targets

TABLE VI
TRANSFER GAPS OF THE POOLED-CONFORMAL POLICY WHEN SHORTAGE IS MORE COSTLY

Train	Test history	Population	n	Matched	Unseen	Gap	95% percentile interval
A	B	Full sample	20,641	0.6831	0.6932	+0.0101	[0.0076, 0.0128]
A	C	Full sample	20,641	0.6831	0.7303	+0.0472	[0.0391, 0.0554]
B	A	Full sample	20,641	0.6940	0.6908	-0.0032	[-0.0059, -0.0006]
B	C	Full sample	20,641	0.6940	0.7306	+0.0365	[0.0275, 0.0447]
C	A	Full sample	20,641	0.7248	0.7120	-0.0128	[-0.0219, -0.0050]
C	B	Full sample	20,641	0.7248	0.7086	-0.0162	[-0.0242, -0.0083]

TABLE VII
DEFINITE-MISS LOWER BOUNDS ON REAL FINAL-WEEK ROWS, OVERALL (WITH 95% PERCENTILE CLUSTER-BOOTSTRAP INTERVALS) AND BY RECORDED STOCKOUT HOURS

Trained on	Population	Rows	Bound	95% interval
A	Real history	35,000	0.0895	[0.0852, 0.0942]
B	Real history	35,000	0.0854	[0.0808, 0.0896]
C	Real history	35,000	0.0724	[0.0684, 0.0766]

Recorded stockout hours	Rows	A	B	C
0 (complete day)	20,641	0.0730	0.0708	0.0579
1-4	4,614	0.1797	0.1708	0.1465
5-8	4,980	0.1283	0.1169	0.1044
9-16	4,765	0.0334	0.0327	0.0298

TABLE VIII
ABLATIONS RECORDED IN RUN E, AS PAIRS OF VALUES IN THE ORDER NAMED (MEAN ORDER COST AT EQUAL COSTS UNLESS THE SCOPE SAYS OTHERWISE)

Recorded diagnostic	Values	Scope
Raw-sales vs. demand-quantile	0.4075 / 0.3984	Equal-cost matched cells
Uncalibrated vs. pooled q90 coverage	0.8121 / 0.8832	Final donor proxy; nominal 0.90
Direct vs. recovered-history TFT	0.4999 / 0.4690	Equal-cost deep subset
Short vs. full lag set	Not available	Not run after the lock
Static vs. adaptive calibration	0.3984 / 0.4039	Equal-cost sensitivity
One-day vs. two-day inventory	0.3984 / 0.2579	Hypothetical carry-over
Contract vs. oracle-weather WAPE	33.48% / 33.40%	Target-day oracle; not deployable

a high one, so a model trained on recorded sales does well when excess stock is the expensive error and falls behind when shortage is. This qualifies the warning that treating sales as demand degrades as censored data accumulate [10]: here the penalty appears when shortage is costly.

The ablations recorded in run E (Table VIII) agree with the locked comparisons: averaged over the matched histories at equal costs, the pooled-conformal demand quantiles cost 0.3984 against 0.4075 for the raw-sales point model with residual quantiles. Pooled calibration moved q90 coverage from 0.8121 to 0.8832 against a nominal 0.90. The adaptive variant cost 0.4039 against 0.3984 for static calibration, so its drift correction did not pay off in this week. A hypothetical two-day carry-over of unsold stock lowered the cost from 0.3984 to 0.2579, which shows how much the cost level depends on the one-day shelf-life assumption. Giving the model the target-day weather and promotion values, which are unknown when the order is placed, changed the q50 WAPE only from 33.48% to 33.40%. A reduced-lag ablation was not run, because fitting a new model after the final week had been scored would fall outside the lock.

The main limitations are the following.

- *Proxy and mechanisms.* Donor-day sales equal demand only on complete days, which depend on demand and stocking, so the proxy does not identify demand on censored days, where we give only a lower bound. Mechanisms A, B, and C are specified simulations, not counterfactual inventory paths; transfer results measure sensitivity to these histories, not transport to another retailer or stocking policy.

- *Populations and neural models.* Kaplan–Meier scores only estimable rows. The neural policies cover the 600-series subset with a small CPU and epoch budget, and the official baseline was reproduced only in part, so they are reference points; a fair comparison needs a full training budget and the same scored rows as the tree-based policies.
- *Cost model.* Rankings depend on the locked cost ratios and on the one-day shelf life.
- *One evaluated week.* Results are stable across its seven dates (Section V-D), but one week cannot show seasonal effects, and any later protocol evaluated on it is exploratory. The 45,000 series outside run E’s sample have never been scored, so a fresh locked replication on them is possible.
- *Intervals and provenance.* The intervals use only 200 bootstrap replicates, do not resample dates separately, and have no multiplicity adjustment. No run hashed the notebook source it executed, and the upstream dataset revision and download date are unknown, so the local file hashes are the operative identifiers. All runs used Linux, Python 3.11.15, two logical CPUs, and 8.42 GB of RAM, run E without a GPU. The released outputs omit large intermediate files, so Phase 5 cannot resume from them without regenerating Phases 0–4.

VII. CONCLUSION

A matched donor-proxy design makes stockout-aware forecasting and ordering comparisons on FreshRetailNet-50K checkable: models trained on differently censored histories are scored against the same outcomes, under a protocol fixed

before the final week was read. Within this design, the value of calibrated demand quantiles depends on the cost ratio. They gave the lowest cost in every train/test cell when shortage was more costly, while a raw-sales point model with residual quantiles was cheaper in every cell when excess stock was more costly. Evaluating the calibrated policy on an unseen mechanism raised its cost by at most 0.0472 normalized units, the smallest worst case among the policies that read the evaluation history, and TimesNet-recovered histories lowered TFT order costs within the deep subset. These findings hold for donor days, the simulated mechanisms, and the locked cost settings. Testing them on censored demand itself requires a new protocol; testing them on the 45,000 unscored series requires only a new lock.

CODE AND DATA AVAILABILITY

The notebooks, the saved run outputs, and the scripts that regenerate every table and figure without retraining any model are available at <https://github.com/Research-maniacs/stockout-aware-demand-forecasting>. The outputs of runs S, P, and E are under `results/`, and run E's locked protocol is in `manifests/final_test_lock.json`. The dataset is not redistributed; it is available from Hugging Face [5] under CC BY 4.0. The local files are identified by the SHA-256 hashes below; the last line is the digest of run E's locked protocol.

```
train.parquet 6706832db892bbae4969c19d87e07975
              d2543d2ba7d7d4756360654785de5a3d
eval.parquet  1b118840664280c6b88bffc84c80eelf
              54c05d911e354b7599e5da10995e960e
lock digest   4ab12c4bddc06b4bb32ba8825353071b
              964ef5bfdd423bd4c876ac382180c1b6
```

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TODO: AI-use disclosure required by IEEE (system used, where, and to what extent) and any funding acknowledgments.

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